Problem Statement: Voice Biomarker Disease Tracking (Audio AI)

The Challenge: Patients with conditions such as asthma, cardiovascular issues, neurological disorders, post-brain stroke, Parkinson's, or depression typically only consult doctors every three months. Consequently, pharmaceutical companies lack continuous data on patient health between visits.

Background:
Patients suffering from chronic conditions such as:

- Asthma
- Cardiovascular diseases
- Neurological disorders
- Post-stroke complications
- Parkinson's disease
- Depression

Typically consult doctors only once every few months. This creates a significant gap in continuous health monitoring, making it difficult for doctors and pharmaceutical companies to:

- Track disease progression
- Monitor treatment effectiveness
- Detect early warning signs

Problem: There is currently **no scalable, non-invasive, and continuous monitoring system** that can track a patient's condition on a daily basis between clinical visits.

Objective:
Build an AI-powered Voice Diary Application that enables patients to:

- Record a 15-second daily voice sample
- Automatically analyze their voice using AI/ML models
- Track changes in health conditions over time

The AI Solution: Develop a daily "Voice Diary" application. The patient speaks into the app for 15 seconds, and the AI analyzes "Vocal Biomarkers" such as tremors, breathlessness, and pitch to track disease progression or the efficacy of prescribed medication.

Participants are required to develop a system that:

- Captures short voice recordings from users
- Extracts **vocal biomarkers**, such as:
    - Voice tremors
    - Breathlessness
    - Pitch variation
    - Speech rate
    - Pause patterns
- Uses AI/ML models to:
    - Detect anomalies
    - Track disease progression
    - Evaluate medication effectiveness

Features to Implement:

- Voice Recording Module (15 sec input)
- Audio Preprocessing (noise reduction, normalization)
- Feature Extraction (MFCC, pitch, frequency, etc.)
- AI/ML Model for analysis
- Progress Tracking Dashboard
- Alert system for abnormal patterns (optional)

Suggested Tech Stack:

Frontend: React.js / Next.js, Flutter (for mobile apps)

Backend: Node.js / Python (Flask / FastAPI / Django)

AI/ML: Python, TensorFlow / PyTorch, Librosa (audio processing), Scikit-learn

Cloud (Optional): Firebase / AWS / Google Cloud

Database: MongoDB / PostgreSQL / Firebase / MySQL

Deployment: AWS / GCP / Docker / Vercel / Render / Railway /Netlify

Expected Deliverables

Participants must submit:

- Working Prototype / Demo
- Source Code (GitHub Repository)
- PPT Presentation (optional)
- Problem Understanding & Approach (Report)
- Demo Video (optional but recommended)

Constraints:

- Voice recording should be short (10–20 seconds)
- Solution should be non-invasive and user-friendly
- Use of pre-trained models is allowed, but customization is encouraged
- **To ensure the final solution is cost-effective and scalable, it should not rely on third-party paid APIs. The students should aim to build their AI solutions using open-source tools or custom models to provide the required vocal biomarker analysis**

Expected Impact:

- Continuous patient monitoring
- Early detection of health deterioration
- Improved treatment effectiveness
- Data-driven healthcare insights

Vocal Biomarkers represent a cutting-edge advancement in AI, enabling accessible and scalable healthcare solutions.

Development Rules:

- All coding must happen during the hackathon window.
- Use of open-source libraries allowed.
- Use of pretrained models allowed.
- Previously built full solutions not allowed.
- Plagiarism leads to disqualification